

SIDDHARTH GHODE

Backend / Full-Stack Software Engineer

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[LinkedIn](#) [GitHub](#) [Portfolio](#) [LeetCode](#)

SUMMARY

Backend-focused Software Engineer with hands-on depth in concurrency-safe transaction design and hybrid recommendation systems – uncommon at entry level. Shipped a 17,500+ line full-stack platform (31 REST endpoints, OTP-secured auth) deployed end-to-end on Render, Supabase, and Docker.

EDUCATION

Master of Computer Science 2024–Present
Savitribai Phule Pune University, Department of Computer Science

Bachelor of Computer Science 2021–2024
Savitribai Phule Pune University

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

Backend: Django, Django REST Framework, REST API Design, JWT, OAuth 2.0, RBAC

Frontend: React, Tailwind CSS, HTML5, CSS3

Databases: PostgreSQL, SQLite

Data / ML: Pandas, NumPy, Scikit-learn, TF-IDF, Cosine Similarity, Collaborative Filtering

DevOps / Deployment: Docker, Docker Compose, Nginx, Unicorn, Render, Supabase, GitHub Actions

Testing: Pytest, Pytest-Django, DRF APIClient

Tools: Git, GitHub, Linux, Postman

Core Concepts: OOP, Concurrency Control, Transactions, Query Optimization, Authentication & Authorization

PROJECTS

University Library Management & Book Recommendation System Nov 2025 – Jul 2026
[Live Demo](#) | [GitHub](#)

Python • Django REST Framework • React • PostgreSQL • Scikit-learn • JWT • OAuth 2.0 • Docker • Render • Supabase

- Architected 31 REST API endpoints across 5 Django apps and 12 relational models, covering authentication, catalog management, borrowing, notifications, and analytics.
- Engineered a hybrid recommendation engine (TF-IDF, 500-feature vocabulary + interaction-weighted collaborative filtering across a top-50 similar-user pool) across a 6,800+ book catalog; architected a multi-tier caching layer with department-scoped keys and automated invalidation on book interactions and catalog imports, with cold-start fallbacks.
- Designed concurrency-safe borrowing workflows using `transaction.atomic()` and `select_for_update()` to eliminate inventory race conditions under concurrent requests.
- Built a 2-stage OTP verification funnel (Argon2-hashed, 60s resend cooldown, 5-attempt lockout, single-use tokens) with JWT access/refresh rotation.
- Deployed a 3-service Docker Compose stack (React, Django, PostgreSQL) on Render/Supabase with Unicorn and Nginx; resolved production CORS and integration issues via deployment-log debugging.
- Authored a 123-test Pytest suite (Pytest-Django, DRF APIClient) at 80% backend coverage across auth, atomic borrowing, and the recommendation engine; automated via a 3-stage GitHub Actions CI pipeline.

PortfolioHub – Full Stack Portfolio Management Platform 2026
[GitHub](#)

Python • Django REST Framework • React • PostgreSQL • Docker

- Built a portfolio management platform with secure CRUD, media uploads, pagination, filtering, and search, exposed through REST APIs in Django REST Framework and PostgreSQL.
- Implemented JWT + OAuth 2.0 authentication with session management and device tracking; containerized the full stack with Docker for consistent deployment.
- Designed a modular React frontend with protected routes, reusable components, and a responsive Tailwind CSS UI integrated against the REST API.

PROBLEM SOLVING & CERTIFICATIONS

59+ problems solved on [LeetCode](#) spanning Arrays, DP, Hash Tables, and Binary Search (Python3, Java, C++).

Data Visualisation – [Certificate](#) **Linux Essentials** – [Certificate](#)